

Concept Review

Ultrasonic Sensors

Why Do We Need Ultrasonic Sensors?

Many mechatronics systems rely on distance sensors for obstacle avoidance, object detection, and localization. Ultrasonic sensors are included within a category of sensors that measure distance. Compared to other distance sensors that work by detecting light, ultrasonic sensors use reflected soundwaves, making them reliable across lighting conditions and for detecting objects with differing light reflectivity.

Ultrasonic Sensors

Ultrasonic sensors operate using signal emission and signal detection to perform distance estimation through the time-of-flight principle. A sound wave at a frequency above the human hearing range is emitted and then reflected by nearby objects with a time delay proportional to the distance of the object. By measuring the amount of time taken for the emitted signal to be detected back at the sensor (the time-of-flight), the distance of objects from the sensor can be estimated. Ultrasonic sensors can come in varying sizes and configurations, and two different ones are shown in Figure 1.



a. HC-SR04

b. Other ultrasonic sensor here

Figure 1: Different ultrasonic sensor configurations

Figure 2 shows an example of a laser-based time of flight sensor which estimates the straight-line distance to a subject of interest.

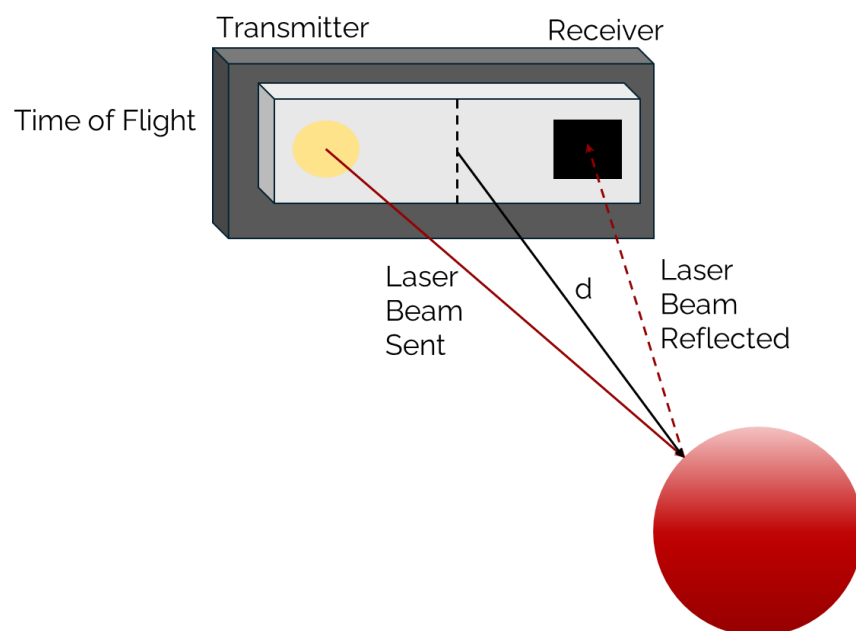


Figure 2: Distance estimate from a laser-based time of flight sensor

Using the time delay from when light is emitted to when it is received, Δt , and the speed of light, c , we can use the following equation to estimate the distance, d , to an object:

$$d = c\Delta t/2 \quad (1)$$

In equation 1, the speed of light c is a constant equal to 299,792,458 m/s. Depending on the sensor manufacturer, this calculation may be done within the sensor directly. As an example, the STM VL53L8CHVoGC will provide distance values directly.

Another interesting characteristic of modern Time-of-Flight sensors is multizone measurements. Multizone refers to a grid like series of measurements which provide more information than a single beam. This information can be useful if you're interested in interpreting the shape of an object within proximity to the Time-of-Flight sensor. Sensors such as the STM VL53L8CHVoGC can provide you with multizone distance values.

Time of Flight sensors also have different technical characteristics based on the manufacturer's specifications. Examples of different Time of Flight sensor technologies:

- VL53L8CHVoGC uses a single photon avalanche diode (SPAD) detection array.
- ADSD3100 uses a Complementary Metal Oxide Semiconductor (CMOS) sensor.

For a VL53L8CHVoGC the SPAD detector array can generate a multizone distance measurement as seen in Figure 3. For this example, a maximum distance of 0.5m is represented by the dark red and a bright blue are values close to the sensor array.

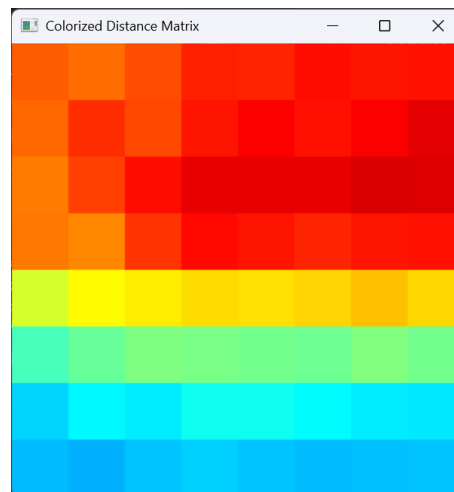


Figure 3: Colorized multizone distance information from VL53L8CHVoGC

Selecting an Ultrasonic Sensor

Datasheets are used to select an appropriate sensor for a specific application and to determine additional design constraints on the rest of the system. For time-of-flight sensors, some key datasheet considerations include the following:

Dynamic range: This describes the full range of operating conditions, including both distance and reflectivity, that the sensor is expected to detect. This value is the ratio of the maximum to the minimum values of each measurement parameter.

Resolution: This refers to the smallest detectable change in distance. Lower resolutions may be sufficient for less critical tasks, while high resolutions are useful for applications that require more detailed measurement, such as imaging.

Field of view: This describes the dimensions of the area within which objects are detectable by the sensor. In sensors containing both an emitter and a receiver, the field of view is determined by the field of view of both components.

Sampling rate: This value indicates how quickly the sensor can output measurements. The minimum sampling rate depends on the response time requirements for the specific application.

Comparison to Other Distance Sensors

The ultrasonic sensor falls within a category of sensors that employ various technologies for distance measurement. The type of sensor that is best suited for an application depends on specific requirements such as cost, range, resolution, accuracy, and speed. The ultrasonic sensor is a cost-effective option for simple object detection. For applications requiring faster readings at a high accuracy, time-of-flight sensors are a more suitable choice but can come at a higher cost. When choosing between a light-based or a sound-based distance sensor, additional considerations include both the target object and operating environment. The performance of ultrasonic sensors is unaffected by certain material properties of the target object such as reflectivity and color, or by environmental factors such as light, dust, and humidity, unlike light-based sensors.



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